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Table 4 shows the changes in the number of HIV RNA particles and helper T cells (T_H cells) in a person infected with HIV who was not receiving any treatment.

Table 4

Time after infection / years	HIV RNA particles per cm^3 of blood	T_H cell count / cells per mm^3 of blood
0	0	1000
1	80 000	305
2	20 000	854
6	22 000	812
10	70 000	597
11	120 000	150
12	135 000	5

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A scientist concluded that other infections are more likely to develop in this person after 11 years.

Use the data in **Table 4** and your own knowledge of the immune response to explain the scientist's conclusion.

[3 marks]



One type of infection is pneumonia, caused by the bacterium *Streptococcus pneumoniae*.

A trainee microbiologist cultured *S. pneumoniae* on an agar plate.

The trainee microbiologist:

- disinfected the laboratory bench
- used a sterile pipette to transfer *S. pneumoniae* from a broth bottle to an agar plate
- took a glass spreader and spread the *S. pneumoniae* across the agar plate
- sealed the lid with four strips of tape.

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Suggest **one** other action the trainee microbiologist took during this method to ensure the technique was aseptic.

Justify why this action was needed.

Do **not** refer to gloves, masks or safety glasses in your answer.

[2 marks]

Action _____

Justification _____

5

Turn over for the next question

Turn over ►

